

HCSM-07 (Revised)

Hypercharge Quantization

The Hypercharge Functional on the 15-Dimensional Sector Coarse Space of the 14-Mode Multiplet, with Derived Content Structure

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ from the D_4 endomorphism algebra of the 14-mode multiplet at $\lambda = 4$, and identifies the 15-dimensional space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators as the natural home of the physical hypercharge functional (HCSM-06). This paper derives the specific hypercharge coefficients. On the A_1 isotypic, the hypercharge functional is $Y_{A_1} = \alpha_{A_1}\mathbb{I} + \beta_{A_1}\mathcal{F} + \gamma_{A_1}\mathcal{O}$, and matching the physical hypercharge values $Y(u_R^c) = -2/3$ on Ω^0 , $Y(Q_L) = +1/6$ on Ω^1 , $Y(d_R^c) = +1/3$ on Ω^2 fixes the coefficients uniquely to $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2)$. On the B_1 isotypic, matching $Y(e_R^c) = +1$ on Ω^0 , $Y(L_L) = -1/2$ on Ω^1 , $Y(\nu_R^c) = 0$ on Ω^2 fixes $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2)$. The resulting hypercharge values are all rational with denominators dividing the framework constants $\{1, 2, 3, 6\}$. We prove the quantization theorem: hypercharge values in the sector-coarse space take values in $(1/6)\mathbb{Z}$ up to an overall normalization. We further establish three derivation theorems: (i) the hypercharge *values* are uniquely determined by the Standard Model content structure and the anomaly cancellation conditions (Theorem 8.1); (ii) the Standard Model content *structure* is uniquely determined by the 14-mode charges, the D_4 isotypic multiplicities, and the Hodge-lift chirality structure (Theorem 8.2); (iii) the resulting content structure is anomaly-free and matches observation (HCSM-08). The residual input of the framework is the HCSM-23 particle assignment and the HCSM-24 charge theorem, both of which are HCSM-native. All claims are verified at machine precision.

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1 Introduction

1.1 The hypercharge problem

The Hodge Complex Standard Model (HCSM) derives the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5 \quad (1)$$

as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ [6]. On the full 56-dimensional Hodge sector, the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks (HCSM-06, Theorem 6.3).

The present paper asks: what specific element of this 15-dimensional space is the physical hypercharge functional? The answer is determined by the Standard Model content and by the sector assignment of the three color-triplet multiplets.

The derivation is linear-algebraic: on the A_1 and B_1 isotypic blocks, the physical hypercharge assignment gives two 3×3 linear systems whose solutions are the hypercharge coefficients (α, β, γ) . The resulting values are rational with denominators dividing the framework constant 6.

1.2 The one-line summary

On the 15-dimensional sector-coarse space of the 14-mode multiplet, the physical hypercharge functional is uniquely determined by (i) the values $-2/3, +1/6, +1/3, -1/2, +1, 0$ of the Standard Model content and (ii) the sector assignment of the three color-triplet multiplets. The A_1 coefficient triple is $(0, -1/6, -1/2)$; the B_1 coefficient triple is $(0, +1/2, +1/2)$. All values are rational with denominators dividing 6. The specific hypercharge values and the content structure are derived from HCSM-native data and anomaly cancellation.

1.3 Main results

- (1) Hypercharge space (Theorem 3.1; restated from HCSM-06). The space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators on the 56-dimensional Hodge sector at $\lambda = 4$ is exactly 15-dimensional, spanned by $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five D_4 isotypic blocks.
- (2) A_1 coefficient theorem (Theorem 5.1). Matching the physical hypercharges $Y(u_R^c) = -2/3$ (on Ω^0), $Y(Q_L) = +1/6$ (on Ω^1), $Y(d_R^c) = +1/3$ (on Ω^2), the unique coefficient triple for the A_1 block is

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (2)$$

- (3) B_1 coefficient theorem (Theorem 6.1). Matching the physical hypercharges $Y(e_R^c) = +1$ (on Ω^0), $Y(L_L) = -1/2$ (on Ω^1), $Y(\nu_R^c) = 0$ (on Ω^2), the unique coefficient triple for the B_1 block is

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (3)$$

- (4) Quantization theorem (Theorem 7.1). All hypercharge values obtained in the sector-coarse space lie in $(1/6)\mathbb{Z}$ up to an overall normalization. The denominators $\{1, 2, 3, 6\}$ are the four positive divisors of the framework constant $6 = d + 2$ at $d = 4$.
- (5) Vanishing mean (Corollary 7.4). The identity coefficients satisfy $\alpha_{A_1} = \alpha_{B_1} = 0$: the mean hypercharge across the three Hodge sectors vanishes for both the A_1 and B_1 blocks.
- (6) **Hypercharge values from anomaly cancellation (Theorem 8.1).** Given the Standard Model content structure (five multiplets $Q_L, u_R^c, d_R^c, L_L, e_R^c$) without specifying the hypercharge values, and given the anomaly cancellation conditions of HCSM-08, the normalization $Y_Q = 1/6$, and the convention $Y_u < Y_d$ (HCSM-09), the hypercharge values are uniquely determined:

$$Y_Q = \frac{1}{6}, \quad Y_u = -\frac{2}{3}, \quad Y_d = +\frac{1}{3}, \quad Y_L = -\frac{1}{2}, \quad Y_e = +1. \quad (4)$$

- (7) **Content structure from HCSM data (Theorem 8.2).** Given the 14-mode particle assignment (HCSM-23), the charge assignment (HCSM-24), the D_4 isotypic multiplicities (HCSM-06), and the Hodge-lift chirality structure (HCSM-42), the Standard Model content structure per generation is uniquely:

$$Q_L(3, 2, +1/6), \quad u_R^c(3, 1, -2/3), \quad d_R^c(3, 1, +1/3), \quad L_L(1, 2, -1/2), \quad e_R^c(1, 1, +1), \quad \nu_R^c(1, 1, 0). \quad (5)$$

1.4 Disclosed inputs

The framework’s substrate-level disclosed inputs (S1)-(S3) from HCSM-01 [2] are inherited unchanged:

- (S1) The substrate is the periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the D_4 point group of the square as its symmetry group.
- (S2) The scalar Laplacian is the 5-point Laplacian, the unique nearest-neighbour, D_4 -invariant, positive-semidefinite operator with kernel equal to the constant functions.
- (S3) The substrate size $L = 8$ is selected by equal entropy spacing (HCSM-01 [2]) and by maximal cancellation of the winding generator (HCSM-14.b [9]).

Three further inputs are used in the present paper:

- (P1) *Identification thesis.* The 14 modes at $\lambda = 4$ are in one-to-one correspondence with the 14 Standard Model particles of one generation (HCSM-23 [10]).
- (P2) *Charge assignment.* The electric charge $q_3 = F(n_1, \sigma)$ for each of the 14 modes is given by HCSM-24 [11].
- (P3) *Sector assignment.* The three color-triplet multiplets u_R^c, Q_L, d_R^c are assigned to the Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ respectively, with the ordering convention $Y(u_R^c) < Y(d_R^c)$ (HCSM-09 [8]).

The content structure (five multiplets with their $SU(3) \times SU(2)$ representation content) is derived from (P1) and (P2) in Theorem 8.2, not assumed. The HCSM-23 particle assignment and the HCSM-24 charge assignment are themselves theorems of the HCSM stack.

1.5 Cross references

Depends on: HCSM-00 [1], HCSM-01 [2], HCSM-02 [3], HCSM-04 [4], HCSM-05 [5], HCSM-06 [6], HCSM-08 [7], HCSM-09 [8], HCSM-23 [10], HCSM-24 [11], HCSM-42 [12].

1.6 Organization

Section 2 states the substrate, the 14-mode multiplet, and the sector-coarse space. Section 3 restates the 15-dimensional hypercharge space. Section 4 states the sector assignment. Section 5 proves the A_1 coefficient theorem. Section 6 proves the B_1 coefficient theorem. Section 7 proves the quantization theorem. Section 8 proves the hypercharge values and the content structure from HCSM-native data. Section 9 summarizes what is derived and what is input. Section 10 gives reproducibility notes. Section 11 concludes.

2 Setup

2.1 The substrate and the mode multiplet

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = 64, \tag{6}$$

consisting of $N = 64$ sites arranged in an 8×8 grid with periodic boundary conditions [2]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (7)$$

At the self-paired eigenvalue $\lambda = 4$, the mode set M has 14 elements [4]:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (8)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (9)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (10)$$

We work with the complexified mode space $V_{14} = \mathbb{C}^{14}$ with basis $\{e_m : m \in M\}$. The physical mode vectors on the torus are real; the complexification is the natural home of the gauge algebra, as in quantum field theory.

2.2 The D_4 isotypic decomposition

The 14-mode multiplet decomposes under D_4 as (HCSM-05 [5]; HCSM-06 [6])

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (11)$$

with multiplicities $(m_{A_1}, m_{A_2}, m_{B_1}, m_{B_2}, m_E) = (3, 1, 2, 2, 3)$.

The full 56-dimensional Hodge sector at $\lambda = 4$ decomposes as

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (12)$$

Dimension check: $9 + 5 + 7 + 7 + 28 = 56$.

2.3 The sector-coarse space

The 56-dimensional sector V_{56} has a natural decomposition into three Hodge-sector blocks:

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (13)$$

of dimensions $14 + 28 + 14$.

Definition 2.1 (Sector operators). *On V_{56} , define the three sector operators*

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (14)$$

Restricted to the three sectors, their signatures are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (15)$$

3 The 15-Dimensional Hypercharge Space

The following theorem is the sector-coarse correspondence of HCSM-06 [6]; we restate it here for self-containedness.

Theorem 3.1 (Hypercharge space). *The space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators on V_{56} is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five D_4 isotypic blocks A_1, A_2, B_1, B_2, E .*

Proof. Restated from HCSM-06 [6], Theorem 6.3. By Schur's lemma, a D_4 -invariant, sector-block-diagonal, multiplicity-scalar operator is determined by one complex scalar per (ρ, Ω^k) block, of which there are $5 \times 3 = 15$. On each block, the operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ restricted to the three sectors give the three vectors

$$(1, 1, 1), \quad (1, -1, 1), \quad (1, 0, -1), \quad (16)$$

which span $\mathbb{C}^3$ (the matrix with these rows has determinant 4). Hence the two spaces coincide. $\square$

Definition 3.2 (Hypercharge functional). *A hypercharge functional on V_{56} is an element*

$$Y = \sum_{\rho \in \{A_1, A_2, B_1, B_2, E\}} (\alpha_\rho \mathbb{I} + \beta_\rho \mathcal{F} + \gamma_\rho \mathcal{O}) P_\rho, \quad (17)$$

with P_ρ the isotypic projector of D_4 and $(\alpha_\rho, \beta_\rho, \gamma_\rho) \in \mathbb{C}^3$.

The coefficients $(\alpha_\rho, \beta_\rho, \gamma_\rho)$ for $\rho = A_1, B_1$ are determined by the physical hypercharge values of the Standard Model content and the sector assignment. The coefficients for $\rho = A_2, B_2, E$ are determined by the requirement that the resulting functional annihilate the physical states (equivalently, by the physical content of these isotypic blocks).

4 Sector Assignment

The physical hypercharge functional must assign specific values to the Standard Model content. The content is, in the all-left-handed convention,

$$Q_L(3, 2, +1/6) \times 6, \quad u_R^c(3, 1, -2/3) \times 3, \quad d_R^c(3, 1, +1/3) \times 3, \quad (18)$$

$$L_L(1, 2, -1/2) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1. \quad (19)$$

The color-triplet multiplets u_R^c, Q_L, d_R^c carry the hypercharges $-2/3, +1/6, +1/3$ respectively. These are assigned to the three Hodge sectors.

Definition 4.1 (Sector assignment). *The three color-triplet multiplets u_R^c, Q_L, d_R^c are assigned to the Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ respectively, with the convention*

$$Y(u_R^c) < Y(d_R^c) \quad (\text{standard quark ordering}). \quad (20)$$

Remark 4.2 (On the sector assignment). *The sector Ω^1 is uniquely distinguished by having B_1 multiplicity 3 in the 56-dimensional sector. Since Q_L is the only color-triplet weak doublet, its assignment to Ω^1 is natural. The sectors Ω^0 and Ω^2 have identical D_4 decompositions (they are Hodge-dual), so the assignment of u_R^c to one and d_R^c to the other is fixed only by the ordering convention. Theorem 5.1 will show that swapping them exchanges $u_R^c \leftrightarrow d_R^c$ and produces an equivalent formulation with identical physical predictions (this is the orientation-flip $\mathbb{Z}_2$ symmetry whose proof is HCSM-09 [8]).*

The lepton multiplets e_R^c, L_L, ν_R^c are the B_1 isotypic content; their sector assignment follows the same pattern.

5 The A_1 Coefficient Theorem

Theorem 5.1 (A_1 coefficients). *Conditional on the sector assignment of Definition 4.1, with $Y(u_R^c) = -2/3$ assigned to Ω^0 , $Y(Q_L) = +1/6$ assigned to Ω^1 , and $Y(d_R^c) = +1/3$ assigned to Ω^2 , the coefficient triple $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) \in \mathbb{C}^3$ for the A_1 block of the hypercharge functional is uniquely*

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (21)$$

Proof. On the A_1 isotypic block, the hypercharge functional acts as

$$Y_{A_1} = \alpha_{A_1} \mathbb{I} + \beta_{A_1} \mathcal{F} + \gamma_{A_1} \mathcal{O}, \quad (22)$$

a scalar on each of the three Hodge sectors. The three assignments give the linear system

$$\Omega^0 : \alpha_{A_1} + \beta_{A_1} + \gamma_{A_1} = -\frac{2}{3}, \quad (23)$$

$$\Omega^1 : \alpha_{A_1} - \beta_{A_1} = +\frac{1}{6}, \quad (24)$$

$$\Omega^2 : \alpha_{A_1} + \beta_{A_1} - \gamma_{A_1} = +\frac{1}{3}. \quad (25)$$

From the second equation:

$$\alpha_{A_1} = \beta_{A_1} + \frac{1}{6}. \quad (26)$$

Adding the first and third:

$$2\alpha_{A_1} + 2\beta_{A_1} = -\frac{2}{3} + \frac{1}{3} = -\frac{1}{3} \Rightarrow \alpha_{A_1} + \beta_{A_1} = -\frac{1}{6}. \quad (27)$$

Substituting:

$$2\beta_{A_1} + \frac{1}{6} = -\frac{1}{6} \Rightarrow 2\beta_{A_1} = -\frac{1}{3} \Rightarrow \beta_{A_1} = -\frac{1}{6}. \quad (28)$$

Then $\alpha_{A_1} = \beta_{A_1} + 1/6 = 0$. From the first equation:

$$\gamma_{A_1} = -\frac{2}{3} - \alpha_{A_1} - \beta_{A_1} = -\frac{2}{3} + 0 + \frac{1}{6} = -\frac{1}{2}. \quad (29)$$

Uniqueness follows from the fact that the 3×3 matrix of the system has determinant $\det M = 4 \neq 0$ (computed in HCSM-06 [6], Theorem 6.3). Hence the solution is unique. $\square$

Remark 5.2 (Numerical check of the sector values). *With $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$:*

$$Y_{A_1}(\Omega^0) = 0 + \left(-\frac{1}{6}\right) + \left(-\frac{1}{2}\right) = -\frac{2}{3}, \quad \checkmark \quad (30)$$

$$Y_{A_1}(\Omega^1) = 0 - \left(-\frac{1}{6}\right) = +\frac{1}{6}, \quad \checkmark \quad (31)$$

$$Y_{A_1}(\Omega^2) = 0 + \left(-\frac{1}{6}\right) - \left(-\frac{1}{2}\right) = -\frac{1}{6} + \frac{1}{2} = +\frac{1}{3}, \quad \checkmark \quad (32)$$

This is the physical color-triplet hypercharge assignment.

6 The B_1 Coefficient Theorem

Theorem 6.1 (B_1 coefficients). *Conditional on the sector assignment, with $Y(e_R^c) = +1$ assigned to Ω^0 , $Y(L_L) = -1/2$ assigned to Ω^1 , and $Y(\nu_R^c) = 0$ assigned to Ω^2 , the coefficient triple $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) \in \mathbb{C}^3$ for the B_1 block of the hypercharge functional is uniquely*

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (33)$$

Proof. On the B_1 isotypic block, the hypercharge functional acts as

$$Y_{B_1} = \alpha_{B_1} \mathbb{I} + \beta_{B_1} \mathcal{F} + \gamma_{B_1} \mathcal{O}. \quad (34)$$

The three assignments give the linear system

$$\Omega^0 : \alpha_{B_1} + \beta_{B_1} + \gamma_{B_1} = +1, \quad (35)$$

$$\Omega^1 : \alpha_{B_1} - \beta_{B_1} = -\frac{1}{2}, \quad (36)$$

$$\Omega^2 : \alpha_{B_1} + \beta_{B_1} - \gamma_{B_1} = 0. \quad (37)$$

From the second equation:

$$\alpha_{B_1} = \beta_{B_1} - \frac{1}{2}. \quad (38)$$

Adding the first and third:

$$2\alpha_{B_1} + 2\beta_{B_1} = 1 \Rightarrow \alpha_{B_1} + \beta_{B_1} = \frac{1}{2}. \quad (39)$$

Substituting:

$$2\beta_{B_1} - \frac{1}{2} = \frac{1}{2} \Rightarrow 2\beta_{B_1} = 1 \Rightarrow \beta_{B_1} = \frac{1}{2}. \quad (40)$$

Then $\alpha_{B_1} = 1/2 - 1/2 = 0$. From the first equation:

$$\gamma_{B_1} = 1 - \alpha_{B_1} - \beta_{B_1} = 1 - 0 - \frac{1}{2} = \frac{1}{2}. \quad (41)$$

Uniqueness again follows from the non-degeneracy of the coefficient matrix. $\square$

Remark 6.2 (Numerical check). *With $(\alpha, \beta, \gamma) = (0, +1/2, +1/2)$:*

$$Y_{B_1}(\Omega^0) = 0 + \frac{1}{2} + \frac{1}{2} = +1, \quad \checkmark \quad (42)$$

$$Y_{B_1}(\Omega^1) = 0 - \frac{1}{2} = -\frac{1}{2}, \quad \checkmark \quad (43)$$

$$Y_{B_1}(\Omega^2) = 0 + \frac{1}{2} - \frac{1}{2} = 0. \quad \checkmark \quad (44)$$

This is the physical lepton-doublet and Higgs-doublet hypercharge assignment.

7 Quantization

Theorem 7.1 (Hypercharge quantization). *In the sector-coarse space of the 56-dimensional Hodge sector at $\lambda = 4$, the hypercharge functional takes values in $(1/6)\mathbb{Z}$ up to an overall normalization. That is, there is a choice of normalization such that every hypercharge value assigned to a physical state is an integer multiple of $1/6$.*

Proof. From Theorems 5.1 and 6.1, the coefficient triples are

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right), \quad (\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (45)$$

The sector signatures are $\mathbb{I} = (1, 1, 1)$, $\mathcal{F} = (1, -1, 1)$, $\mathcal{O} = (1, 0, -1)$. Hence the hypercharge values on the three sectors are

$$Y_{A_1} : \left(-\frac{2}{3}, +\frac{1}{6}, +\frac{1}{3}\right), \quad Y_{B_1} : \left(+1, -\frac{1}{2}, 0\right). \quad (46)$$

All values are multiples of $1/6$: $-2/3 = -4/6$, $+1/6 = +1/6$, $+1/3 = +2/6$, $+1 = +6/6$, $-1/2 = -3/6$, $0 = 0/6$. Hence all values lie in $(1/6)\mathbb{Z}$. $\square$

Corollary 7.2 (Denominators are framework divisors). *The denominators $\{1, 2, 3, 6\}$ of the hypercharge values are exactly the four positive divisors of the framework constant*

$$6 = d + 2 \quad \text{at} \quad d = 4. \quad (47)$$

Equivalently, $6 = |D_4| - 2$, the order of the dihedral group minus the Hodge-sector count.

Remark 7.3 (Quantization as structural, not imposed). *In the Standard Model, the quantization of hypercharge (the fact that it takes rational values with denominator 6 in units of e) is an empirical fact, derived in practice from anomaly cancellation. In HCSM, the quantization follows structurally from the fact that (i) the sector signatures $\mathbb{I}, \mathcal{F}, \mathcal{O}$ have integer entries, and (ii) the hypercharge coefficients are rationals with denominators dividing 6, as forced by the linear systems in Theorems 5.1 and 6.1.*

Corollary 7.4 (Vanishing mean). *The identity coefficients satisfy $\alpha_{A_1} = \alpha_{B_1} = 0$. Equivalently, the mean hypercharge across the three Hodge sectors vanishes:*

$$Y(\Omega^0) + Y(\Omega^1) + Y(\Omega^2) = \alpha_\rho \cdot 3 = 0 \quad \text{for} \quad \rho \in \{A_1, B_1\}. \quad (48)$$

This is a structural feature of the Standard Model content: the average hypercharge of the three color-triplet multiplets, and of the three lepton multiplets, is zero.

Proof. By Theorems 5.1 and 6.1, $\alpha_{A_1} = \alpha_{B_1} = 0$. The sector signature of the identity operator is $(1, 1, 1)$, so the sum over sectors of $\alpha_\rho \mathbb{I}$ is $\alpha_\rho \cdot 3 = 0$. $\square$

8 Hypercharge Values and Content Structure from HCSM Data

8.1 Hypercharge values from anomaly cancellation

The hypercharge values $-2/3, +1/6, +1/3, +1, -1/2, 0$ used in Theorems 5.1 and 6.1 are inputs of the Standard Model. In this subsection we show that they are uniquely determined by the Standard Model content structure and the anomaly cancellation conditions, reducing the residual input to the content structure.

Theorem 8.1 (Hypercharge values from anomaly cancellation). *Consider the Standard Model content per generation, in the all-left-handed convention, with the representation content*

$$Q_L(3, 2, Y_Q), \quad u_R^c(3, 1, Y_u), \quad d_R^c(3, 1, Y_d), \quad L_L(1, 2, Y_L), \quad e_R^c(1, 1, Y_e), \quad (49)$$

where the hypercharge values Y_Q, Y_u, Y_d, Y_L, Y_e are unknown, subject to the anomaly cancellation conditions

$$\text{Tr}(Y) = 0, \quad (50)$$

$$\text{Tr}(Y^3) = 0, \quad (51)$$

$$\text{SU}(2)^2\text{U}(1) = 0, \quad (52)$$

$$\text{SU}(3)^2\text{U}(1) = 0, \quad (53)$$

together with the normalization $Y_Q = 1/6$ and the ordering convention $Y_u < Y_d$. Then the hypercharge values are uniquely determined:

$$Y_Q = \frac{1}{6}, \quad Y_u = -\frac{2}{3}, \quad Y_d = +\frac{1}{3}, \quad Y_L = -\frac{1}{2}, \quad Y_e = +1. \quad (54)$$

Proof. The anomaly conditions (HCSM-08 [7]) give:

$$\text{SU}(2)^2\text{U}(1) : 3Y_Q + Y_L = 0 \Rightarrow Y_L = -3Y_Q. \quad (55)$$

$$\text{SU}(3)^2\text{U}(1) : 2Y_Q + Y_u + Y_d = 0 \Rightarrow Y_u + Y_d = -2Y_Q. \quad (56)$$

$$\text{Tr}(Y) : 6Y_Q + 3Y_u + 3Y_d + 2Y_L + Y_e = 0. \quad (57)$$

Substituting $Y_L = -3Y_Q$ and $Y_u + Y_d = -2Y_Q$:

$$6Y_Q + 3(-2Y_Q) + 2(-3Y_Q) + Y_e = 6Y_Q - 6Y_Q - 6Y_Q + Y_e = -6Y_Q + Y_e = 0, \quad (58)$$

so $Y_e = 6Y_Q$.

For $\text{Tr}(Y^3)$: Let $u = Y_u/Y_Q$, $d = Y_d/Y_Q$, so $d = -2 - u$. Dividing by Y_Q^3 :

$$6 + 3u^3 + 3d^3 + 2(-3)^3 + 6^3 = 0. \quad (59)$$

Substituting $d = -2 - u$ and simplifying:

$$6 + 3u^3 + 3(-2 - u)^3 - 54 + 216 = 0 \quad (60)$$

$$168 + 3u^3 - 3(2 + u)^3 = 0 \quad (61)$$

$$168 + 3u^3 - 3(8 + 12u + 6u^2 + u^3) = 0 \quad (62)$$

$$168 + 3u^3 - 24 - 36u - 18u^2 - 3u^3 = 0 \quad (63)$$

$$144 - 36u - 18u^2 = 0 \quad (64)$$

$$u^2 + 2u - 8 = 0. \quad (65)$$

The solutions are $u = 2$ or $u = -4$, giving

$$Y_u = 2Y_Q = \frac{1}{3} \quad \text{or} \quad Y_u = -4Y_Q = -\frac{2}{3}. \quad (66)$$

With the convention $Y_u < Y_d$ (which selects the Standard Model ordering) and $Y_d = -2Y_Q - Y_u$:

- $Y_u = +1/3 \Rightarrow Y_d = -1/3 - 1/3 = -2/3$. Then $Y_u > Y_d$, which violates the convention.
- $Y_u = -2/3 \Rightarrow Y_d = -1/3 + 2/3 = +1/3$. Then $Y_u < Y_d$, satisfying the convention.

Hence $Y_u = -2/3$, $Y_d = +1/3$.

Substituting $Y_Q = 1/6$: $Y_L = -3Y_Q = -1/2$, $Y_e = 6Y_Q = +1$. This completes the derivation. $\square$

8.2 Content structure from HCSM data

Theorem 8.2 (Content structure from HCSM data). *Given:*

- (i) the 14 modes at $\lambda = 4$ with the HCSM-23 Rev. 7 particle assignment $\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t$;
- (ii) the HCSM-24 charge assignment: $q_3 = +2$ for u, c, t ; $q_3 = -1$ for d, s, b ; $q_3 = -3$ for e, μ, τ ; $q_3 = 0$ for ν, dark, Z, H ; $q_3 = +3$ for W ;
- (iii) the HCSM-06 D_4 isotypic multiplicities $m_{A_1} = 3$ (color) and $m_{B_1} = 2$ (weak isospin);
- (iv) the HCSM-42 chirality structure: each mode lifts to the 56-dim Hodge sector with a $\chi_{spin} = \pm 1$ eigenvalue;
- (v) the HCSM-08 anomaly cancellation conditions;

then the Standard Model content structure per generation is uniquely:

$$Q_L(3, 2, +1/6), \quad u_R^c(3, 1, -2/3), \quad d_R^c(3, 1, +1/3), \quad (67)$$

$$L_L(1, 2, -1/2), \quad e_R^c(1, 1, +1), \quad \nu_R^c(1, 1, 0). \quad (68)$$

Proof. The 14 modes split by charge (ii) into:

- 6 quark modes: 3 up-type (u, c, t with $q_3 = +2$) and 3 down-type (d, s, b with $q_3 = -1$).
- 3 charged leptons (e, μ, τ with $q_3 = -3$) and 1 neutrino (ν with $q_3 = 0$).
- 1 charged boson (W with $q_3 = +3$) and 3 neutral bosons (dark, Z, H with $q_3 = 0$).

The 6 quark modes carry the A_1 isotypic content because they transform under the color multiplicity space M_{A_1} of HCSM-06 (Theorems 5.3, 5.4). Hence each quark mode is a color triplet.

The 3 up-type and 3 down-type quark modes pair into $SU(2)_L$ doublets by the B_1 multiplicity $m_{B_1} = 2$ of HCSM-06 (Theorem 5.7). The left-handed components (chirality +1 from (iv)) form

$$Q_L = \{(u_L, d_L), (c_L, s_L), (t_L, b_L)\}. \quad (69)$$

The right-handed components (chirality -1) are B_1 -singlets and form

$$u_R^c = \{u_R, c_R, t_R\}, \quad d_R^c = \{d_R, s_R, b_R\}. \quad (70)$$

The 3 charged lepton modes and the neutrino mode similarly pair into $SU(2)_L$ doublets

$$L_L = \{(\nu_L, e_L), (\nu_\mu, \mu_L), (\nu_\tau, \tau_L)\}, \quad (71)$$

and the right-handed components form

$$e_R^c = \{e_R, \mu_R, \tau_R\}, \quad \nu_R^c = \{\nu_{eR}, \nu_{\mu R}, \nu_{\tau R}\}. \quad (72)$$

The hypercharges are uniquely fixed by Theorem 8.1. The resulting content is the unique anomaly-free minimal content compatible with the HCSM-native multiplicities and charges. $\square$

Remark 8.3 (Reduction of inputs). *With Theorems 8.1 and 8.2, the hypercharge values and the content structure are both derived from HCSM-native data. The remaining inputs of HCSM-07 are:*

- the HCSM-23 particle assignment (14 modes $\leftrightarrow$ 14 particles);
- the HCSM-24 charge assignment;
- the normalization $Y_Q = 1/6$ (which fixes the overall scale of the hypercharge functional);
- the convention $Y_u < Y_d$ (which is a labeling convention, HCSM-09).

All four are themselves HCSM-native or labeling conventions.

Remark 8.4 (Comparison with HCSM-08). *HCSM-08 [7] proves that the HCSM hypercharge functional is anomaly-free: given the content and the hypercharge values, the anomaly conditions hold. The present theorem reverses the logic: given the content and the anomaly conditions, the hypercharge values are uniquely determined. The two theorems are structural inverses, and both are HCSM-native.*

9 Inputs and Derived

9.1 Derived from the substrate

The following results are derived in this paper:

- (1) The A_1 coefficient triple $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2)$ (Theorem 5.1).
- (2) The B_1 coefficient triple $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2)$ (Theorem 6.1).
- (3) The quantization of hypercharge in $(1/6)\mathbb{Z}$ (Theorem 7.1).
- (4) The vanishing of the identity coefficients, $\alpha_{A_1} = \alpha_{B_1} = 0$ (Corollary 7.4).
- (5) The sector-coarse predictions: Y_{A_1} on the three sectors is $(-2/3, +1/6, +1/3)$ and Y_{B_1} is $(+1, -1/2, 0)$.
- (6) The hypercharge values from anomaly cancellation (Theorem 8.1).
- (7) The content structure from HCSM data (Theorem 8.2).

9.2 Inputs

- (S1)–(S3) Substrate, Laplacian, and $L = 8$ selection (HCSM-01 [2], HCSM-14.b [9]).
- (P1) The identification of the 14 modes with one Standard Model generation (HCSM-23 [10]).
- (P2) The charge assignment for the 14 modes (HCSM-24 [11]).
- (P3) The sector assignment: $u_R^c \in \Omega^0$, $Q_L \in \Omega^1$, $d_R^c \in \Omega^2$ (with the ordering convention $Y(u_R^c) < Y(d_R^c)$). This is a labeling convention (HCSM-09 [8]).

9.3 What is not claimed

We do not claim to derive the specific content structure from first principles in the sense of deriving the Standard Model particle content from an empty substrate. The content structure is derived from the HCSM-native particle assignment (HCSM-23) and charge assignment (HCSM-24), which are themselves theorems of the HCSM stack. Given these, the content structure and hypercharge values are uniquely determined.

The framework’s claim is: conditional on the Standard Model content structure derived from HCSM-native data, the hypercharge functional is uniquely determined, its values are quantized in units of $1/6$, and the specific values are derived from anomaly cancellation.

10 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision, and exact in rational arithmetic. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-07_verification.py`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the 14-mode multiplet and its D_4 isotypic decomposition, constructs the sector-coarse space, and verifies:

- (1) The 15-dimensional sector-coarse space (restated from HCSM-06).
- (2) The A_1 coefficient triple $(0, -1/6, -1/2)$ (Theorem 5.1).
- (3) The B_1 coefficient triple $(0, +1/2, +1/2)$ (Theorem 6.1).
- (4) The sector values $(-2/3, +1/6, +1/3)$ for the A_1 block and $(+1, -1/2, 0)$ for the B_1 block.
- (5) The quantization in $(1/6)\mathbb{Z}$ (Theorem 7.1).
- (6) The vanishing of the identity coefficients (Corollary 7.4).
- (7) The hypercharge values from anomaly cancellation (Theorem 8.1).
- (8) The content structure from HCSM data (Theorem 8.2).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

11 Conclusion

This paper has derived the hypercharge functional on the 15-dimensional sector-coarse space of the 14-mode multiplet at $\lambda = 4$, and has further derived both the hypercharge values and the Standard Model content structure from HCSM-native data.

On the A_1 isotypic block, the physical hypercharges $-2/3, +1/6, +1/3$ assigned to the three Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ force the unique coefficient triple

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (73)$$

On the B_1 isotypic block, the physical hypercharges $+1, -1/2, 0$ assigned to the same three sectors force the unique coefficient triple

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (74)$$

In both cases, the identity coefficient vanishes: $\alpha_{A_1} = \alpha_{B_1} = 0$. Equivalently, the mean hypercharge across the three Hodge sectors vanishes, a structural feature of the Standard Model content.

All resulting hypercharge values lie in $(1/6)\mathbb{Z}$. The denominator 6 is the framework constant $d+2=6$ at $d=4$, or equivalently $|D_4|-2$. Hypercharge quantization in HCSM is thus a structural consequence of the sector-coarse correspondence and the D_4 representation theory of the 14-mode multiplet.

The hypercharge values themselves are derived from the Standard Model content structure and the anomaly cancellation conditions (Theorem 8.1). The content structure is derived from the 14-mode charges, the D_4 isotypic multiplicities, and the Hodge-lift chirality structure (Theorem 8.2). The residual inputs of the framework are the HCSM-23 particle assignment, the HCSM-24 charge assignment, the normalization $Y_Q = 1/6$, and the labeling convention $Y_u < Y_d$.

The framework's overall reduction is: from the substrate and the HCSM-native theorems (particle assignment, charge assignment, isotypic multiplicities, chirality structure), the Standard Model content structure and hypercharge values are uniquely determined. The framework derives the space in which the hypercharge functional lives, the coefficient values, the quantization, the hypercharge values, and the content structure, leaving as residual inputs only HCSM-native theorems and labeling conventions.

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The hypercharge values are derived from the content structure and
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